

# JORDAN NEWMAN

## Systems Software Engineer | Embedded Software Engineer

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### SUMMARY

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Systems software engineer with 20+ years of experience spanning kernel development, operating systems, networking, security, embedded firmware, and large-scale production infrastructure. Built and operated ISP/CDN infrastructure scaling to 3,000-4,000 servers, 10 CDN locations, and 65+ Gbps peak traffic. Developed production FreeBSD kernel modules, DDoS mitigation systems, high-performance C networking software, x86 bootloaders, an educational OS kernel, and embedded AVR firmware.

### TECHNICAL SKILLS

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**Languages:** C, C++, Go, Python, C#, Objective-C, PHP, Perl, JavaScript, Node.js, Shell, Visual Basic

**Systems:** FreeBSD, Linux, kernel modules, system calls, bootloaders, operating systems, kqueue, GDB, TCP/IP

**Networking/Security:** DDoS detection/mitigation, packet capture, DNS analysis, libpcap, IPFW, authentication, telemetry

**Embedded:** Embedded C, Atmel AVR, SPI, UART, I2C, Arduino, circuit design, soldering, hardware troubleshooting

**Cloud/DevOps:** Azure DevOps, Docker, Kubernetes, Terraform, CI/CD, security scanning, deployment automation

**Databases/Web:** MySQL, MultiValue/Universe, OAuth/OAuth2, SMTP OAuth, React, jQuery, PHP, web services

### SELECTED TECHNICAL HIGHLIGHTS

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- Built and operated infrastructure spanning 3,000-4,000 servers, 10 CDN locations, and 65+ Gbps peak traffic.
- Developed PrimeHTTPD in C around FreeBSD kqueue; supported 150,000+ concurrent connections across ~200 production servers.
- Created FreeBSD kernel modules for security, filesystem virtualization, and filesystem observability in customer-facing production systems.
- Developed libpcap-based network and DNS analyzers for real-time traffic visibility, DDoS analysis, and DNS recursion attack detection.
- Developed embedded C firmware for battery-powered wireless Atmel AVR smart smoke/CO2 detector prototypes using SPI, UART, and I2C.
- Built custom x86 OS kernel and bootloader with protected mode, IDT, heap allocation, port I/O, VGA output, and GDB debugging.

### SELECTED SYSTEMS PROJECTS

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**TAFOS Operating System:** Developed educational x86 OS kernel in C and assembly with custom MBR bootloader, protected mode, IDT, heap allocator, port I/O, VGA output, and GDB support.

**KeepClean:** Developed FreeBSD kernel module hooking syscalls to monitor execution activity, protect system assets, and report telemetry to userspace services.

**VirtualDir:** Developed FreeBSD kernel module remapping filesystem paths through syscall interception; deployed customer-facing across approximately 30 production servers over 5-8 years.

**StatCache:** Developed FreeBSD kernel module instrumenting stat() and lstat() activity for filesystem visibility and performance analysis.

**PrimeDump:** Developed libpcap-based traffic analyzer decoding Ethernet, IP, TCP, UDP, and ICMPv6 headers with ncurses real-time views and IPFW integration.

**PrimeDNSTop:** Developed DNS traffic monitor parsing UDP/53 and RFC 1035 queries to detect recursion attacks and rank source/domain activity.

**AVR Smart Smoke/CO2 Detector:** Developed contract firmware in embedded C for battery-powered wireless Atmel AVR prototypes using SPI, UART, and I2C interfaces.

### PROFESSIONAL EXPERIENCE

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**Advantive - Senior Development Specialist** | Remote / Tampa Bay, FL | July 2022 - Present

- Implemented hands-on Azure DevOps, Docker, Kubernetes, Terraform, CI/CD, and security-scanning workflows across engineering teams.
- Partnered with architecture teams to improve deployment reliability, DevOps policies, and production software delivery practices.
- Developed internal AI workflows, reusable libraries, and engineering documentation to improve team productivity and consistency.

**DDI Systems - Senior Development Specialist** | Manalapan, NJ | November 2021 - July 2022

- Modernized legacy VB.NET software by porting components to C#, replacing deprecated and insecure implementations.
- Integrated Microsoft Office 365 SMTP OAuth, improving authentication security for customer-facing software.

**ISPrime LLC - CEO** | Weehawken, NJ | April 2018 - January 2020

- Led datacenter modernization, operational cleanup, and software platform improvements for production hosting infrastructure.
- Refined internally developed monitoring, security, and infrastructure tools to improve operational efficiency.

**DDI Systems - Computer Programmer** | Manalapan, NJ | April 2017 - April 2018

- Integrated ERP software with SellerCloud and Essendant systems, automating order, catalog, image, and SDS data workflows.

- Developed customer-facing import tools for MultiValue/Universe databases and SFTP-based supplier integrations.

**MFCXY, Inc. - Computer Programmer** | *Chicago, IL* | *December 2014 - September 2015*

- Developed backend server features and Windows client integrations while identifying performance and security issues.
- Built middleware to migrate data across database platforms and improve operational continuity.

**ISPrime Inc. - CIO / Partner** | *Weehawken, NJ* | *March 2001 - December 2014*

- Built and operated infrastructure spanning 3,000-4,000 servers across 10 CDN locations and multiple datacenters.
- Sustained peak traffic exceeding 65 Gbps while supporting mission-critical customer workloads.
- Developed PrimeHTTPD, a high-performance C web server optimized around FreeBSD kqueue that supported 150,000+ concurrent connections and demonstrated performance competitive with, and in some workloads exceeding, nginx.
- Deployed PrimeHTTPD across approximately 200 production servers supporting high-traffic customer environments.
- Developed DDoS detection and mitigation systems protecting infrastructure during attacks involving up to 750,000 compromised hosts.
- Created PrimeDump and PrimeDNSTop, libpcap-based network analysis tools for traffic anomalies and DNS attacks.
- Developed production FreeBSD kernel modules including KeepClean, VirtualDir, and StatCache for security and filesystem observability.
- Built monitoring, telemetry, authentication, and security systems used across thousands of production servers.
- Automated provisioning, deployment, and infrastructure management across globally distributed hosting platforms.

**FatWallet.com - Systems Administrator** | *Wisconsin / Remote* | *December 2000 - June 2001*

- Rebuilt production infrastructure to resolve performance and stability issues without customer-facing downtime.
- Built automated alerting systems for website availability, server health, and operational response.

## EDUCATION

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**Brookdale Community College** - Computer Science coursework; left during first semester to pursue ISPrime full-time during rapid company growth.